

Weekly training log, week 9, updated with all EfficientNet trials

Week 9

Module	EEEM068, Applied Machine Learning
Project	Industrial waste classification on WaRP
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Focus	EfficientNetV2 experiments on WaRP C

1. Experimental settings

Dataset	WaRP C
Classes	28
Train and test split	8805 train images, 1583 test images in the project log. Evaluation notebooks report 49 test batches and 1551 labelled test samples in the classification report.
Input size	224 x 224
Main metrics	Top 1 accuracy and macro F1. Extra evaluation also reports precision, recall, macro AUC and mAP.
Goal	Compare EfficientNetV2 variants and training recipes as CNN baselines for WaRP C.

2. Trial overview

Trial	Notebook and setup	Backbone	Training recipe	Best top 1	Macro F1	Comment
1	Original log and paper setup	EfficientNetV2 M, ImageNet pretrained	40 epochs, all layers unfrozen from start, AdamW, ReduceLROnPlateau, CrossEntropy without smoothing or weights	76.60%	74.9%	Best overall result. Use this number in the main comparison table.
2	Simple augmentation baseline	EfficientNetV2 S, ImageNet 1K pretrained	60 epochs. 20 head only and 40 full fine tuning. Differential learning rate, EMA, class weights and label smoothing in phase 1, label smoothing in phase 2	43.97%	41.67%	Overfits strongly after unfreezing. Train accuracy reaches about 99%, while EMA test accuracy plateaus near 44%.
3	MixUp trial	EfficientNetV2 S, ImageNet 1K pretrained	Same two phase setup as Trial 2, but MixUp alpha 0.4 is applied in phase 2	53.19%	51.68%	MixUp improves generalisation over Trial 2 by 9.22 percentage points top 1, but remains below EfficientNetV2 M.

3. Hyperparameters by trial

Item	Trial 1: EfficientNetV2 M paper setup	Trial 2: EfficientNetV2 S simple augmentation	Trial 3: EfficientNetV2 S MixUp
Epochs	40, single phase	60 total. phase 1 equals 20, phase 2 equals 40	60 total. phase 1 equals 20, phase 2 equals 40
Frozen and unfrozen	All layers unfrozen from start	Phase 1 head only. Phase 2 full fine tuning	Phase 1 head only. Phase 2 full fine tuning
Learning rate	0.001 initially. Reduced to 0.0005 at epoch 37	Head LR 0.001. Backbone LR 0.0001 in phase 2	Head LR 0.001. Backbone LR 0.0001 in phase 2
Optimizer	AdamW	AdamW	AdamW
Weight decay	0	0.0001	0.0001
Scheduler	ReduceLROnPlateau, factor 0.3, patience 5	CosineAnnealingLR per phase	CosineAnnealingLR per phase
Loss	CrossEntropy, no label smoothing, no class weights	Phase 1: CrossEntropy with class weights and label smoothing 0.1. Phase 2: label smoothing 0.1	Same as Trial 2, with MixUp soft label loss in phase 2
Sampler and augmentation	Uniform shuffle. No sampler and no minority augmentation	WeightedRandomSampler and minority augmentation. Pretrained CNN pipeline	WeightedRandomSampler and minority augmentation. MixUp alpha 0.4 in phase 2
EMA	Off	On, evaluated using EMA model	On, evaluated using EMA model
Parameters	52,894,224	20,213,356	20,213,356

4. Results summary

Metric	Trial 1: EfficientNetV2 M	Trial 2: Simple augmentation	Trial 3: MixUp
Best or final accuracy	76.60%	43.97%	53.19%
Macro F1	74.92% or 0.749	41.67%	51.68%
Precision	76.72%	64.18%	67.42%
Recall	74.32%	43.88%	51.14%
Macro AUC	98.47%	94.68%	96.52%
mAP	82.23%	60.69%	67.57%

Ranking by top 1 accuracy: EfficientNetV2 M paper setup at 76.60%, followed by EfficientNetV2 S MixUp at 53.19%, followed by EfficientNetV2 S simple augmentation at 43.97%.

Key interpretation: MixUp helped the EfficientNetV2 S recipe compared with simple augmentation, but the EfficientNetV2 M single phase paper setup remained much stronger. The simple augmentation trial shows a clear train and test gap after unfreezing, so it should be discussed as an overfitting or recipe mismatch result rather than as the main EfficientNet baseline.

5. Epoch checkpoints

Trial 1: EfficientNetV2 M paper setup

Epoch	Loss	Train	Test	Best
1	2.571	25.3%	35.4%	35.4%
5	1.758	45.2%	54.2%	54.2%
10	1.525	52.0%	60.9%	60.9%
15	1.393	56.3%	64.0%	66.0%
20	1.268	59.1%	68.1%	68.2%
25	1.186	61.8%	65.8%	68.8%
30	1.110	64.1%	68.5%	69.6%
35	1.064	65.6%	69.7%	70.6%
38	0.866	71.7%	73.9%	73.9%
39	0.843	72.4%	76.6%	76.6%
40	0.781	73.9%	73.5%	76.6%

Trial 2: EfficientNetV2 S simple augmentation

Epoch	Loss	Train	Test EMA	Best
1	2.839	23.1%	3.9%	3.9%
10	2.274	39.7%	17.0%	17.0%
20	2.236	41.6%	25.2%	25.2%
30	0.875	92.7%	36.3%	36.3%
40	0.732	97.5%	42.7%	42.7%
46	0.706	98.4%	43.7%	43.7%
52	0.682	99.1%	44.0%	44.0%
60	0.679	99.2%	42.9%	44.0%

Trial 3: EfficientNetV2 S MixUp

Epoch	Loss	Train	Test EMA	Best
1	2.839	23.1%	3.9%	3.9%
10	2.274	39.7%	17.0%	17.0%

20	2.236	41.6%	25.2%	25.2%
30	1.105	43.7%	34.6%	34.6%
40	0.947	51.9%	46.7%	46.7%
45	0.845	54.5%	50.2%	50.2%
52	0.863	55.4%	52.8%	52.8%
57	0.835	50.8%	53.2%	53.2%
60	0.882	49.6%	52.4%	53.2%

6. Final observations for the report

1. Trial 1 is the reportable EfficientNetV2 M baseline. It reached 76.60% top 1 accuracy and 74.92% macro F1, with strong macro AUC at 98.47% and mAP at 82.23%.
2. Trial 2 confirms that simply adding a two phase pretrained EfficientNetV2 S recipe did not generalise well on WaRP C. The model fitted the training data very strongly but stopped improving on the test set around 44%.
3. Trial 3 shows that MixUp regularisation reduced the overfitting pattern and increased test accuracy from 43.97% to 53.19%, but it was still not competitive with the EfficientNetV2 M paper setup.
4. For the final comparison table, use 76.60% as the EfficientNet result. The other two trials can be included in an appendix or ablation note to show that MixUp helped but the larger EfficientNetV2 M setup was the strongest EfficientNet configuration tested.

7. Short wording to include in the report

We evaluated three EfficientNet based trials on WaRP C. The strongest setup was EfficientNetV2 M trained for 40 epochs with all layers unfrozen from the start, reaching 76.60% top 1 accuracy and 74.92% macro F1. A two phase EfficientNetV2 S recipe with simple augmentation underperformed, reaching 43.97% accuracy and showing a large train and test gap. Adding MixUp in the second phase improved the EfficientNetV2 S result to 53.19%, suggesting better regularisation, but it remained below the EfficientNetV2 M configuration. Therefore, 76.60% is used as the final EfficientNet result in the main model comparison, while the EfficientNetV2 S trials are treated as ablation experiments.